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Digital Logic

Five modules : 4-6 Marks / Easy Subject.

Module 1, Boolean Algebra and Minimization.

- There are two types of systems : Analogous i.e. Continuous
Digital i.e. Discrete.
- We (in CS) don't care about how Gates are actually made, we only care about their behaviour.
- Boolean Algebra : Given by George boole in 1843
 - ↳ Two valued System (0 & 1)
 - ↳ Inspired from prop. logic.
 - ↳ Given by Socrates, Aristotle
- After almost 100 years of boolean alg., Shannon, used it to model the Digital Systems.
- Every data that can be stored in comp. can be repr. as 0 & 1. and this is how actually stored in RAM. RAM contain capacitors which when charged repr. 1, else repr. 0.
- Physically we assign a range of charge like (4V to 5V) to high (i.e. 1) else (0V to 1V) to low (i.e. 0). and transition occurs b/w 1V to 4V.

$A=12 \xrightarrow{\text{Compiler}} A=1100 \rightarrow \text{In memory}$

↓
Program

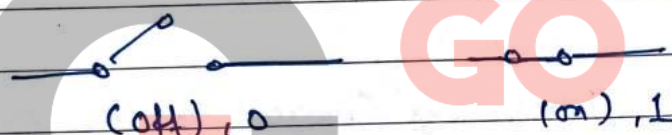


Series of transistors repr. 12

→ Boolean Algebra $\Rightarrow$ Switching Algebra

→ Collection of n bits have 2^n possible states.

→ Binary switch has two states



→ We use boolean variables, like X or Y , to repr. i/p or o/p of a switching circuit. Each var. can either take 0 or 1.

If physical entity's input signal is low then bool var. is 0, and if it's high then it's 1. These symbols out do not have any numeric value, they are just symbols.

→ Binary v/s. Boolean In boolean alg. 0 & 1 don't have a numeric value; they just repr. diff states of circuit. Like in gates 0 means low voltage & 1 repr. range of high voltages. In switches 0 means open switch, and 1 means close circuit.

Binary is used to repr. a numerical value. We can add, divide, etc). It's a no. system.

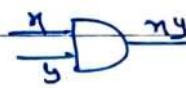
→ Bit $\Rightarrow$ Binary Digits i.e. 0 & 1.

→ Boolean Algebra $\equiv$ Propositional Logic.

↳ Basic operations are (AND, OR, Not)
+ . -



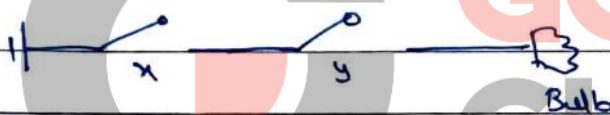
Not Gate



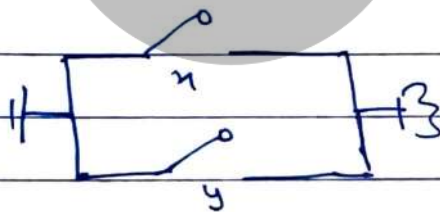
AND Gate



OR Gate



→ AND using switches



→ OR using switches

→ Boolean expression contains 0, 1, $x, y, \dots$, +, ., $\bar{x}$

boolean vars

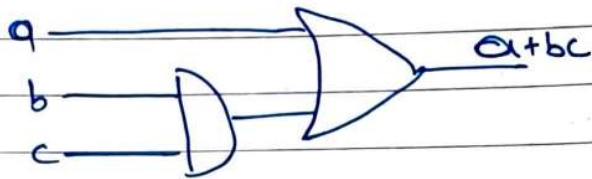
- ① 0 & 1
- ② Boolean variables
- ③ If x & y are bool. exp then
 $\bar{x}, \bar{y}, x+y, x \cdot y, (x), \dots$

→ Each boolean expression corresponds directly to a circuit of logic gates.

Inverter $\Rightarrow$ Not Gate

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Ex: $a+bc$ correspond to

→ Truth Table (also known as table of combinations) specifies the values of a boolean expression for every possible combination of values of variables in the expression.

→ Precedence NOT > AND > OR.

So $a+bc$ means $a+(bc)$ & not $(a+b) \cdot c$

→ There are four more standard boolean operations:

a) NAND ($\uparrow$)

$a \uparrow b = \overline{a \cdot b}$ (Not of And)

b) NOR ($\downarrow$)

$a \downarrow b = \overline{a + b}$ (Not of OR)

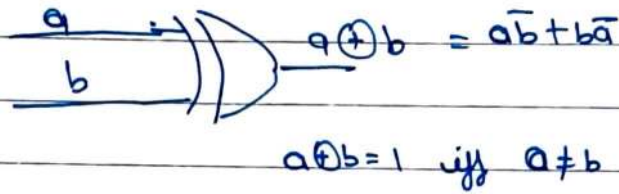
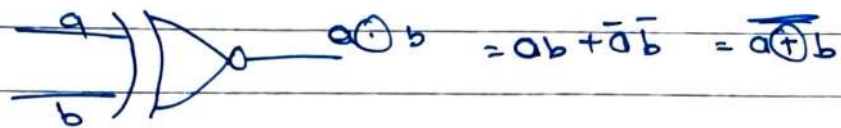
(NAND & NOR gates are universal gates i.e., they alone are functionally complete)

a	b	$\bar{b}$	ab	$a+b$	$a \uparrow b$	$a \downarrow b$
0	0	1	0	0	1	1
0	1	0	0	1	1	0
1	0	x	0	1	1	0
1	1	x	1	1	0	0

Truth table

c) Exclusive OR (EXOR) ($\oplus$)

Also called
Equivalence Operator
↑

d) Ex NOR ($\odot$)

a	b	$a \oplus b$	$a \odot b$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

→ When we arrange truth table such that values of all variables in decimal system is sorted in inc. order.

00	00
01	10
10	11
11	01

Not in std. form.

Standard

→ AND & OR gate can be repr. for many inputs as $a \cdot b \cdot c$ & $a + b + c$ respectively. (Not writing truth table)

* If $\exp 1 = \exp 2$
 then $(\exp 1)^d = (\exp 2)^d$

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NOTE $f = \bar{f}$ is not at all (NEVER) possible.
 So $f \neq \bar{f}$

* Boolean Algebra Laws: Same (isomorphic) to that of PL, Set, Group theory, Bool. lattice.

→ Let C, A & B be any boolean expression:

→ Domination / Annulment Law:

$$A + 1 = 1$$

$$A \cdot 0 = 0 \text{ (Dud of above)}$$

(other in this)

$$A + 1 = \bar{A}$$

$$A + 0 = A$$

$$A \cdot 1 = \bar{A}$$

$$A \cdot 0 = \bar{A}$$

→ Identity Law:

$$A + 0 = A$$

$$A \cdot 1 = A$$

→ Idempotent Law:

$$A + A = A$$

$$A \cdot A = A$$

$$A + \bar{A} = 1$$

$$A \cdot \bar{A} = 0$$

$$A + A = 1$$

$$A \cdot \bar{A} = 0$$

→ Complement Law:

$$A + \bar{A} = 1$$

$$A \cdot \bar{A} = 0$$

→ Double Comp:

$$\bar{\bar{A}} = A$$

→ Commutative Law:

$$A + B = B + A$$

$$A \cdot B = B \cdot A$$

Holds in all op.
 except imp-lication

→ Associative Law:

$$(a+b)+c = a+(b+c)$$

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

Holds in all op.
 except imp-lication

(Can be proved by taking cases)

So

$$(A+B)+C \neq A+(B+C)$$

$$(A \cdot B) \cdot C \neq A \cdot (B \cdot C)$$

→ De Morgan's Law:

$$\overline{a+b} = \bar{a} \cdot \bar{b}$$

$$\overline{a \cdot b} = \bar{a} + \bar{b}$$

"

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→ Distributive Law:

$$a(b+c) = ab+ac$$

$$a+(b \cdot c) = (a+b)(a+c)$$

NOTE AND, OR are not complement of each other
as if $f=ab$ then $\bar{f} \neq a+b$

→ Complementing Boolean Expression using Demorgan's Law: If bool. exp. only

have literals, AND, OR, NOT then demorgan's law can be use to find complement of this boolean expression.

$$\begin{array}{ccccccc} f(a, b, \bar{a}, \bar{b}, +, \cdot, 0, 1) \\ \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \\ \bar{f}(\bar{a}, \bar{b}, a, b, \cdot, +, 1, 0) \end{array}$$

$$\begin{array}{lcl} f = \overline{(x+y) \cdot c} & , & \overline{(x+y) \cdot c} \\ \downarrow & & \downarrow \\ \bar{f} = (x+y) \cdot \bar{c} & , & (\bar{x} \cdot \bar{y}) + \bar{c} \end{array}$$

Simplify $\bar{F}$:

$$\begin{aligned} & \overline{A\bar{B}C} + \overline{ACD} + \overline{BC} \\ &= \overline{A\bar{B}C} \cdot \overline{ACD} \cdot \overline{BC} \\ &= A\bar{B}C \cdot ACD \cdot (\bar{B} + \bar{C}) \\ &= A\bar{B}CD(\bar{B} + \bar{C}) \\ &= \underline{\underline{A\bar{B}CD}} \end{aligned}$$

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$$F = (\overline{w} + x)y + \overline{z}$$

$$\overline{F} = ?$$

$$\begin{aligned} \Rightarrow \overline{F} &= \overline{(\overline{w} + x)y + \overline{z}} \\ &= \overline{[(\overline{w} + x) + \overline{y}]} \cdot \overline{\overline{z}} \\ &= \overline{[(\overline{w} \cdot x) + \overline{y}]} \cdot z \\ &= \overline{[(\overline{w} + \overline{x}) \cdot \overline{y}]} \cdot z \\ &= (\overline{w} + \overline{\overline{w}} + \overline{\overline{x}} + \overline{\overline{y}}) \cdot z \\ &= (\overline{w} + w + \overline{x} + \overline{y}) \cdot z \\ &= (\overline{w} + x + \overline{y}) \cdot z \end{aligned}$$

* $\Rightarrow$ Some more useful (really useful properties)

a) Absorption Law:

i) $\underline{a + ab = a}$ * ✓

ii) $\underline{a \cdot (a + b) = a}$ (Dual of above eq.)

most imp. (*
iii) $\underline{a + \overline{a}b = a + b}$ * ✓

b) Consensus Property:

~~AB + AC + BC~~ $\underline{AB + \overline{A}C + BC = AB + \overline{A}C}$ **

Remember as B is with good of A & C is with bad of A to turn B & C are doing conspiracy being together, so remove them.

⇒ Boolean algebra laws are given in a pair to show that boolean algebra satisfies duality
i.e. if an identity is true
then dual of it will be true

Ex: $a + ba = a$
 $a \cdot (b + a) = a$ (Dual of above identity)

⇒ Dual of a function can be given if bool. exp. only has OR, AND, literal, NOT, then:

$$\begin{array}{ccccccc} f & (a, b, \bar{a}, \bar{b}, \cdot, +, 0, 1) \\ \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \\ f^d & (a, b, \bar{a}, \bar{b}, +, \cdot, 1, 0) \end{array}$$

Ex: $f: a + b + \bar{c} \cdot \bar{d}$
 $f^d = a \cdot b \cdot (\bar{c} + \bar{d})$
 $\bar{f} = \bar{a} \cdot \bar{b} \cdot (c + d)$
 $\bar{f}^d = a + b + c + d$

So, we do not complement the literal while finding the dual of any boolean expression.

NOTE: For some function $f = f^d$ is possible to be true but not for all functions.

Ex: $f = a$, $f^d = a$,

Such fun. are called self dual functions

$f = \bar{f}$ is not at all possible

$\exists f, f = f^d$ & $\forall g, g \neq f^d$

Ques: If $f^d = \bar{f}$ is possible?

⇒ Yes possible, for ex $f = a\bar{b} + b\bar{a}$ i.e. $a \oplus b$.

$$\bar{f} = (\bar{a} + b) \cdot (a + \bar{b})$$

$$f^d = (a + \bar{b}) \cdot (\bar{a} + b)$$

$$\text{So } \bar{f} = f^d$$

→ If $E = F$, then $\bar{E} = \bar{F}$ is

if $E = F$ then $E^d = F^d$ is also true.

Can be used to prove theorem directly.

Some practise:

$$X + \bar{X}B = X + B$$

$$X + X + X + X + X + X = X \quad \rightarrow \text{Absorption Law}$$

a) $\bar{A} (\bar{A}BCD)$

$$= \bar{A} (\bar{A} + \bar{B} + \bar{C} + \bar{D})$$

$$= \bar{A} + \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{D}$$

$$= \underline{\bar{A}} \quad (\text{Absorption law})$$

c) $B\bar{C}DE + B\bar{C}(\bar{D}\bar{E}) + (\bar{B}\bar{C})DE$

$$= B\bar{C}(DE + \bar{D}\bar{E}) + (\bar{B}\bar{C})DE$$

$$= B\bar{C} + \bar{B}\bar{C}DE$$

$$= \underline{B\bar{C} + DE}$$

b) $A(\bar{A} + B) + AB$

$$= A\bar{A} + AB + AB$$

$$= 0 + A(B + \bar{B})$$

$$= \underline{A}$$

d) $\bar{X}YZ + \bar{X}Y\bar{Z} + XZ$

$$= Z(\bar{X}Y + X) + \bar{X}Y\bar{Z}$$

$$= Z(Y + X) + \bar{X}Y\bar{Z}$$

$$= YZ + XZ + \bar{X}Y\bar{Z}$$

$$= Y(Z + \bar{Z}\bar{X}) + XZ$$

$$= YZ + Y\bar{X} + XZ$$

$$= \underline{Y\bar{X} + XZ}$$

(Consensus law)

e) $A\bar{B}C + \bar{A}B\bar{C} + \bar{A}\bar{B}C$

$$= A\bar{B}C + \bar{A}C(B + \bar{B})$$

$$= A\bar{B}C + \bar{A}C$$

$$= C(A\bar{B} + \bar{A}) \quad (\text{Absorption law})$$

$$= C(\bar{B} + \bar{A})$$

$$= \underline{\bar{B}C + \bar{A}C}$$

⇒ Simplification is imp. bcs simplified bool. exp. will give simple circuit from gates. (i.e. less gates required).

⇒ Literal count is a measure that tells us size of bool. exp.
Ex: $\bar{a}\bar{b}c + ab + \bar{c}b + \bar{b}\bar{c} + \bar{a}b\bar{c}$ has 12 literals in 3 variables.

⇒ By simplification/minimization we mean to reduce #literals.

Ques: Prove: $(a+b) \cdot (a+\bar{a}\bar{b}) \cdot c + (\bar{a} \cdot (b+\bar{c})) + \bar{a} \cdot b + abc = a+b+c$

⇒ Case I: $a=1$

$$\begin{aligned} \text{LHS} &= (1+b) \cdot (1+\bar{1}\bar{b}) \cdot c + (0 \cdot (b+\bar{c})) + 0 \cdot b + 1 \cdot bc \\ &= 1 \cdot 1 \cdot c + (0) + 0 + 0 \\ &= 1 + 1 + 0 + 0 = \underline{\underline{1}} \end{aligned}$$

$$\text{RHS} = 1 + b + c = \underline{\underline{1}}$$

Case II: $a=0$

$$\begin{aligned} \text{LHS} &= (0+b) \cdot (0+1\bar{b}) \cdot c + (1 \cdot (b+\bar{c})) + 1 \cdot b + 0 \cdot bc \\ &= b \cdot \bar{b}c + \bar{b}\bar{c} + 1 \\ &= 0 + \bar{b}\bar{c} + b \\ &= (\bar{b} \cdot \bar{c}) + b \\ &= b + \bar{b}c \\ &= \underline{\underline{b+c}} \end{aligned}$$

$$\text{RHS} = 0 + b + c = \underline{\underline{b+c}}$$

∴ Hence Proved.

⇒ Can also be done using boolean simplification.

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NOTE Cancellation laws do not hold in boolean algebra.

Ex: $\cancel{a} \cdot 1 = \cancel{a} \cdot 0$ } Can't cancel
 $\Rightarrow 1 \neq 0$

Some things related to XOR:

$$\overline{a \oplus b} = a \odot b \quad \overline{\bar{a} \oplus b} = a \oplus \bar{b} = \bar{a} \odot b$$

$$a \oplus b = \overline{a \odot b} = \bar{a} \oplus \bar{b} = \bar{a} \odot b = a \odot \bar{b}$$



Here 1 is diff.



Here also 1 is diff.

Que: which of the following circuits are equivalent?

a)



b)



c)



d)



$$\Rightarrow a = d \text{ and } b = c$$

→ # boolean functions possible.

a) for 1 var.

x	f1	f2	f3	f4
0	0	0	1	1
1	0	1	0	1

So 4 functions are possible

b) for 'n' variable

c) for 0 variables

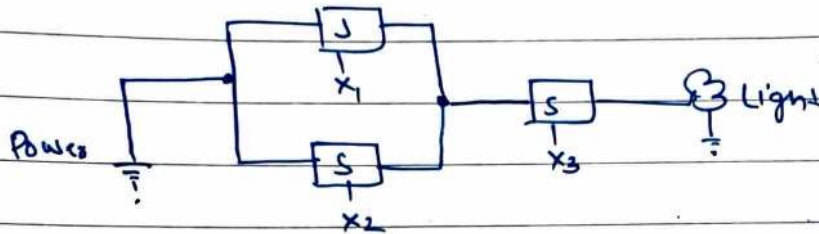
⇒ Two re. always const.

(2^n) functions possible.

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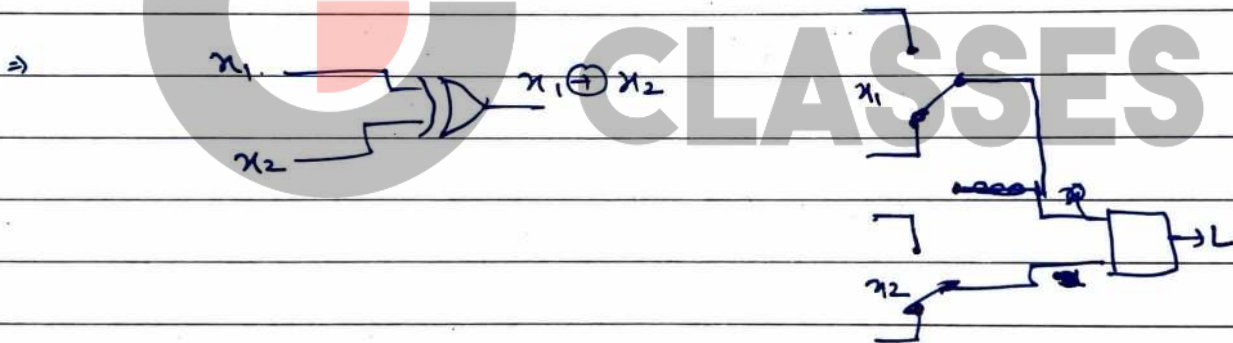
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Ques: What will be logical behaviour of below circuit?



$\Rightarrow (x_1 + x_2) x_3$

Ques: Design circuit for staircase light control. using 2-way switch.
(Both switch off/on, turn off else on)



★ Standard forms of boolean expressions:

- There are many ways to express a boolean expression, it's useful to have a standard or canonical way. It can be derived by truth table. (It might not be simplest form).
- Over 2 var, we can only have 16 (2^2) functions, but some functions can be written in various ways if some std. repr. is not followed.

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There are two principle standard forms:

a) Sum-of-product (SOP)

b) Product-of-sums (POS)

Let's understand them:

→ Product term: AND of literals, or, a single literal.

Ex: $AB, \bar{A}, B\bar{C}$ are product term, but $a+b, \bar{a}\bar{b}$ are not.

→ Sum term: OR of literals, or, a single literal

Ex: $a+b, a, \bar{b}, \bar{a}+b$ are sum term, but $ab, \bar{a}\bar{b}, \bar{a}+\bar{b}$ are not.

→ SOP: Sum/OR of product terms

Ex: $P_1 + P_2 + P_3$ Ex: $a+b+\bar{c}$

→ POS: Product/AND of sum terms

Ex: $(S_1)(S_2)(S_3)$ Ex: $(a+b)(\bar{b})(\bar{a}+c)$

→ we are talking about forms of same exp. like $(a+b)(a+c)$, both are same exp. logically, but, in diff. forms.

Ex: $abc, a+b+c$ are both SOP & POS.

Ques: Is following term in SOP, if not convert.

$$\bar{A}B\bar{C}\bar{D} + A\bar{B}D + \bar{C}\bar{D} + A\bar{D}$$

Ans: Not in SOP bcz of $\bar{A}B\bar{C}\bar{D}$

$$(\bar{A}+\bar{B})(\bar{C} + A\bar{B}D + \bar{C}\bar{D} + A\bar{D})$$

$$\bar{A}\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}D + \bar{C}\bar{D} + A\bar{D}$$

→ Now in SOP.

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→ Minterms: A product term of n literal in which every variable must appear exactly once in original or complemented form.

→ Maxterm: A sum term of n literal in which every variable must appear exactly once in any form.

Ex: if (a, b) then:

	Product term	Sum term	Minterm	Maxterm
a				
a	✓	✓	x	x
$a+b$	x	✓	x	✓
$a\bar{b}$	✓	x	✓	x
$a+b$	x	✓	x	✓
$\bar{a}+a+b$	x	✓	x	x

for 1 variable we have 2 minterms & 2 maxterms. (a & $\bar{a}$)

for 3 variable we have 2^3 i.e. 8 minterms & 8 maxterms

$abc, a\bar{b}c, a\bar{b}\bar{c}, \bar{a}bc$ $a+b+c, a+b+\bar{c}, a+\bar{b}+c, a+\bar{b}+\bar{c}$
 $\bar{a}bc, \bar{a}\bar{b}c, \bar{a}\bar{b}\bar{c}, a\bar{b}\bar{c}$ $\bar{a}+b+c, \bar{a}+\bar{b}+c, \bar{a}+b+\bar{c}, \bar{a}+\bar{b}+\bar{c}$

→ In all of the definitions above SOP, POS, sumterm, minterm..., order of literals does not matter.

→ For ' n ' variables we have 2^n minterms & maxterms possible.

→ Every minterm is '1' i.e. True for exactly one row.

Ex:

a	b	ab	$a\bar{b}$	$\bar{a}b$	$\bar{a}\bar{b}$
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

} i.e. only 1 row.

So, we can say ab is minterm associated with row $a=1, b=1$ & $a\bar{b}$ is minterm associated with row $a=1, b=0$.

Ex: $f(a, b, c)$

a	b	c	associated minterm	associated maxterm
0	0	0	$\bar{a}\bar{b}\bar{c}$ (m_0)	$a+b+c$ (M_0)
0	0	1	$\bar{a}\bar{b}c$ (m_1)	$a+b+\bar{c}$ (M_1)
0	1	0	$\bar{a}b\bar{c}$ (m_2)	$a+\bar{b}+c$ (M_2)
0	1	1	$\bar{a}bc$ (m_3)	$a+\bar{b}+\bar{c}$ (M_3)
1	0	0	$a\bar{b}\bar{c}$ (m_4)	$\bar{a}+b+c$ (M_4)
1	0	1	$a\bar{b}c$ (m_5)	$\bar{a}+b+\bar{c}$ (M_5)
1	1	0	$ab\bar{c}$ (m_6)	$\bar{a}+\bar{b}+c$ (M_6)
1	1	1	abc (m_7)	$\bar{a}+\bar{b}+\bar{c}$ (M_7)

Short Notation of minterms
Short notation of maxterms

→ Similarly, every maxterm is 0, for exactly 1 row.

Ex: $f(a, b)$

a	b	$a+b$	$a+\bar{b}$	$\bar{a}+b$	$\bar{a}+\bar{b}$
0	0	0	1	1	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	1	0

So, we can also see associated maxterms in above table.

→ Also called index

For short notation of minterm (m_i) or maxterm (M_i), fixed order of variables is considered, (alphabetical if not given)

$0 \rightarrow$ minterm $\rightarrow$ ~~Not~~ Complemented
 $1 \rightarrow$ maxterm $\rightarrow$ Not Complemented
 $0 \rightarrow$ minterm $\rightarrow$ Not Complemented
 $1 \rightarrow$ maxterm $\rightarrow$ Complemented.

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$\Rightarrow$ Relationship b/w minterm & maxterm (m_i & M_i) is that both are complement of each other.

Ex: $m_2 = x \cdot \bar{y}$, $M_2 = \bar{x} + y$
 $\bar{m}_2 = \overline{x \cdot \bar{y}} = \bar{x} + y = M_2$
 so $\bar{m}_2 = M_2$ & $\bar{M}_2 = m_2$

M_i is complement of m_i .

NOTE

for a given row of truth table, the minterm is formed by including x_i if $x_i = 1$ & by inc. $\bar{x}_i$ if $x_i = 0$ and the maxterm is formed by including x_i if $x_i = 0$ & by inc. $\bar{x}_i$ if $x_i = 1$.

Ex: $\bar{a}\bar{b}\bar{c}d\bar{e}$ & $\bar{a}+b+c+\bar{d}+e$ are associated with row $(a=0, b=0, c=0, d=1, e=0)$ i.e. row 18. (m_{18}/M_{18})

Ques:

for n variables, how many minterms are there?

$\Rightarrow$ Exactly 2^n (Not almost, misconception: that minterm f^n value is 1).

$\Rightarrow$ Expressing boolean functions using minterms/maxterms. Any bool. fun. is True for some minterms & false for some minterms (Row). Similarly True for some maxterms & false for some maxterms.

- All the minterms having value 1 are called true minterms. (1)
- " " " " " 0 " " false " (0)
- " " maxterms " " 1 " " true maxterms (1)
- " " " " " 0 " " false " (0)

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Ex:

x	y	f ₁
0	0	0
0	1	1
1	0	0
1	1	1

→ f₁ is 0 for m₀→ f₁ is 1 for m₁1 minterms (m₁) or maxterms (m₀ & m₂)0 minterms or maxterms (m₀ & m₂)Imp.

Any boolean function f can be expressed as a summation function.
 (OR) of those minterms for which $f=1$. OR can be expressed as a product (AND) of those maxterms for which $f=0$.

$f = \sum_m (f=1)$ (OR) list of 1-minterms	or	$f = \prod_M (f=0)$ (AND) list of 0-maxterms
--	----	--

So above $f_1 = \bar{x}y + xy$ or $f_1 = (x+y)(\bar{x}+y)$

→ Note that minterms are not 1-minterm, each row is a minterm i.e. a product term having all variables. However only 1-minterms are used to express f .

→ Minterms of function are those minterms which are used to express function i.e. when $f=1$. Maxterm of function are those maxterm when $f=0$.

So, function's minterms & maxterms are mutually ex. & exclusive.

So if i say m_i, m_j, m_k are minterms of $f(a,b,c)$ then m_i, m_j, m_k are 1 and remaining minterms are 0 for $f(a,b,c)$.

f_1	a	b	c	f_1	f_2
	0	0	0	0	0
	0	0	1	1	0
	0	1	0	0	0
	0	1	1	0	1
	1	0	0	1	0
	1	0	1	0	1
	1	1	0	0	1
	1	1	1	1	1

Canonical SOPs

$$\begin{cases} f_1 = \sum m(1, 4, 7) = \bar{x}\bar{y}z + \bar{x}y\bar{z} + xyz \\ f_2 = \sum m(3, 5, 6, 7) = \bar{x}yz + x\bar{y}z + xy\bar{z} + xyz \end{cases}$$

Minterms of f_1 are 1, 4, 6, minterms of f_1 are 0, 2, 3, 5, 6
 " " " f_2 " 3, 5, 6, 7. " " f_2 " 0, 1, 2, 4

Canonical POSs

$$\begin{cases} f_1 = \prod M(0, 2, 3, 5, 6) = (\bar{x}\bar{y} + \bar{z})(x + \bar{y}z)(x + y + \bar{z})(\bar{x} + y + \bar{z}) \\ f_2 = \prod M(0, 1, 2, 4) = (\bar{x}\bar{y} + \bar{z})(x + y + \bar{z})(x + \bar{y} + \bar{z})(\bar{x} + y + \bar{z}) \end{cases}$$

⇒ Canonical SOPs: Each fun. can be repr. by CSOPs by ORing all minterms of that function i.e. Sum of minterms = Canonical SOP.

⇒ Canonical POSs: Each fun. can be repr by CPOSs by ANDing all maxterms of that function
Product of maxterms = Canonical POS. (0 Maxterms)

→ for any boolean fun. we have unique CSOP & CPOS. It can have many SOPs & POSs. (Not unique)

Ex:

a	b	f
0	0	0
0	1	1
1	0	1
1	1	1

$$CSOP = \bar{a}b + a\bar{b} + ab$$

$$f = a + b$$

↳ This is SOP but NOT CSOP.

SOP → Sum of Product Terms (may not have all literals)

CSOP → Sum of minterms

→ $f = \bar{f}$ is never possible.

→ Is CSOP complement of CPOS? Not at all. → CSOP & CPOS both are repr. same func, then how can they be complements.

→ Is CSOP dual of CPOS? Not at all.

→ CSOP & CPOS are equivalent & just principle unique repr. of a function.

Ex: $f = a + b$

CSOP of $f = \bar{a}b + a\bar{b} + ab$ — ①

CPOS of $f = a + b$ — ②

Dual of $f = a \cdot b$ — ③

Comp. of $f = \bar{a} \cdot \bar{b}$ — ④

1 $\equiv$ 2, 1 $\neq$ 3, 1 $\neq$ 4, 2 $\neq$ 3, 2 $\neq$ 4, 3 $\neq$ 4.

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→ In propositional logic, SOP is disj. of Conj. i.e. Disjunctive Normal form (DNF) & POS is Conj. Normal form (CNF)

Boolean Algebra

~~SOP~~ POS~~POS~~ SOP~~CDNF~~ CPOS~~CPOS~~ CSOP

Propositional Logic

CNF

DNF

CCNF (Principle CNF)

CDNF (Principle DNF)

$$\Rightarrow F = \sum_m (F=1) = \prod_M (F=0)$$

$$\bar{F} = \sum_m (\bar{F}=1) = \sum_m (F=0) = \prod_M (\bar{F}=0) = \prod_M (F=1)$$

(used in que)

* → for F, if minterm indices of F = S to all indices in N then maxterm indices of $\bar{F}$ = N-S.

$$\text{Ex: } f(a,b,c) = \sum (1,5,6) = \prod (0,2,3,4,7)$$

$\downarrow$ Minterms $\downarrow$ Maxterms

Ques: Express the boolean function $F = A + \bar{B}C$ as sum of minterms

→ M₁:

$$A \cdot 1 \cdot 1 + \bar{B} \cdot C \cdot 1$$

$$A(B + \bar{B})(C + \bar{C}) + \bar{B}C(A + \bar{A})$$

$$ABC + A\bar{B}C + AB\bar{C} + A\bar{B}\bar{C} + \bar{A}BC + \bar{A}\bar{B}C$$

$$= ABC + A\bar{B}C + AB\bar{C} + A\bar{B}\bar{C} + \bar{A}BC + \bar{A}\bar{B}C$$

$$= \sum (1,4,5,6,7)$$

My Pref

Use any method

M₂:

A means, $\bar{B}C$ means

1--	--01
100,	101,
111,	001
101,	
110,	

$$= \sum (1,4,5,6,7)$$

Ques: Express $F = xy + \bar{x}z$ as product of maxterms

MT: find $f = \sum m(1, 3, 6, 7)$ by prev. method

& then $f = \prod M(0, 2, 4, 5)$

So, best method is find minterms first & then write other terms as maxterms & then POS.

Ques: Express the complement of the following function in CSOP:

a) $f(A, B, C, D) = \sum(0, 2, 6, 11, 13, 14)$

$\Rightarrow \bar{f} = \sum(1, 3, 4, 5, 7, 8, 9, 10, 12, 15)$

b) $f(x, y, z) = \prod(0, 5, 6, 7)$

$\bar{f} = \sum(0, 3, 6, 7)$

★ Minimization of Boolean Expression using K-map

Boole. exp. can gen. be simplified by using the algeb. laws as done previously. However it has ^{many} ~~some~~ problems:

- No ^{easy} systematic procedure / algorithm.
- Hard to tell if it is min or can be simplified further.
- Result depends on order in which we apply boolean law. to minimize.
- d) Error prone.

for ex: $f = \bar{a}\bar{b} + \bar{a}\bar{b}c + \bar{a}b\bar{c} + a\bar{b}c + ab\bar{c} + abc$

$f = \bar{a}\bar{b} + \bar{b}c + b\bar{c} + ab$

↳ Irredundant i.e. not minimum nor can be further minimised.

$f = \bar{a}\bar{b} + b\bar{c} + ac$

↳ minimum

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Do we use Karnaugh Map (KMap) to get minimum size expression from a function. It provides systematic approach by providing visuals to combine terms to get minimum expression. Kmap can give us minimum SOP or minimum POS.

Minimum SOP (or POS) has minimum # terms & each term has minimum # literals. It is not necessarily unique. we may get min. SOP/POS with diff terms but same literal count in them.

* K-map: It can be thought of a special version of truth table designed to ^{ease} the minimization w/o having to use boolean algebra theorems & equation manipulations.

K-map can be used to minimize expression with upto 4 variables easily.

→ Idea behind k-map:

function $f(a,b,c) \rightarrow$ unique SOP

Combine minterms $\leftarrow$ Basically so minterms
 $\hookrightarrow$ Minimum Expression

Que. Which minterms can be combined?

$$a) m_4, m_6 = a\bar{b}\bar{c} + ab\bar{c} \\ = a\bar{c}$$

$$b) m_4, m_7 = \underline{abc + a\bar{b}c} \\ \hookrightarrow \text{Can't be combined.}$$

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→ m_i & m_j can only be combined if only one variable is changed.

Ex $abc + \bar{a}bc = bc$ (only a is changed)

→ In Kmap, truth table is represented in such a way (in matrix/map format) that adjacent cells differ only by one variable.
↳ NOT Diagonally

Ex: $f(a,b,c)$

ab	00	01	11	10
c	0	1	1	0
0	0	1	1	0
1	1	1	1	0

only c changing.

only b changing

Ex2: $f(a,b,c,d)$

ab	00	01	11	10
00	m_0	m_4	m_5	m_1
01	m_2	m_6	m_7	m_3
11	m_8	m_{12}	m_{13}	m_9
10	m_{10}	m_{14}	m_{15}	m_{11}

So, f can be repr. as K-map / SOP / POS / Truth table, etc.
↳ just another repr. of any function.

std. Truth table

In STT, table adjacent rows don't differ by 1 variable (Ex 01 & 10)

→ Each cell in K-Map $\equiv$ a minterm $\equiv$ a maxterm

K-map is thought of 3D i.e. folded i.e. edges are touching i.e.

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→ Adjacent cells: Two cells are adjacent iff they differ exactly by a single variable.

Ex:

	00	01	11	10
0	0	2	5	4
1	1	3	7	6

0, 2 are adjacent 0 & 3 are not adjacent.
4, 0 are adjacent 5 & 0 are not.

→ Cube in k-map: Group of 2^m cells so that every cell in that group is adjacent to exactly m cells in that group.

Ex:

	00	01	11	10
0	0	2	5	4
1	1	3	7	6

Cube of size 2^1

1 is adj. to 3 & 4

3 is adj. to 1

0, 4 is also a cube of size 2

Cube of size 2^2

6 is adj. to 4, 5, 7, (i.e. 3 cells)

4 " " " 6 & 5,

5 " " " 7 & 4, &

7 " " " 5 & 6.

{0, 1, 4, 5}

Cube of size 2^2

{0, 2, 6, 4, 1, 3, 7, 5}

→ Cube of size 8, i.e. each cell is adj. to 3 cells.

0 is adj. to 1, 4, 5.

6 " " " to 2, 4, 7

5 " " " " 4, 7, 1.

You know cubes. now

Ques: Why 2^m cells in a cube in K-map?

→ Because we want minimum SOP. i.e. One product term with x literals cover 2^m minterms where x is # var. in fun.

Ex: $f(a, b, c)$.

We have product term ab , minterms abc, abc or abc, abc product term $\bar{a}$ then minterms $\{abc, abc, abc, abc\} \Rightarrow 2^2$

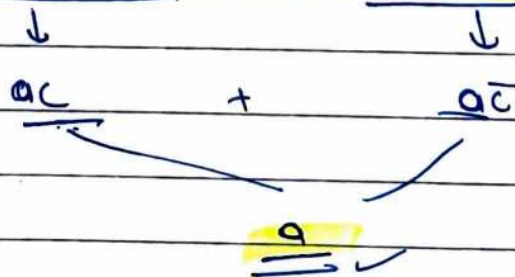
(Similar case for sum term also).

So, this is why we cover cubes of size 2^m & check only those var. which are not changing in cube.

Idea

2 minterms can be reduced to 1 product term iff they change by single variable. (2 minterms can't be reduced to 1 product term). 4, 8 minterms can be reduced to 1 product term iff all minterms have 1 unchanged variable.

Ex: $abc + abc + abc + abc$ (a is unchanged)



→ This product term is covering 4 minterms of $f(a, b, c)$ has 3 var.

So, we create cube of size 2^m so that we can write product term (or sum term) for that cube, & then write min. exp.

Size of cube $\uparrow$ literals in product (or sum) term $\downarrow$

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m.s.o.p. $\Rightarrow$ minimum s.o.p.

for $\rightarrow$ Writing product terms from cubes to find min s.o.p, we make cubes of 1-minterms,

finding m.s.o.p:

a) make k-map.

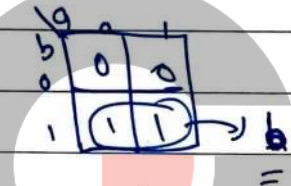
b) ~~make~~ make as large as possible cubes where fun is 1. $\rightarrow$ can overlap

c) ~~make~~ Cover all 1's by min. (max sized) cube.

d) write product terms of all cubes we found above.

e) Sum these terms & it's m.s.o.p.

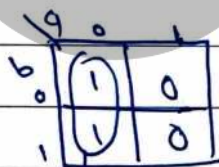
Ex:



b

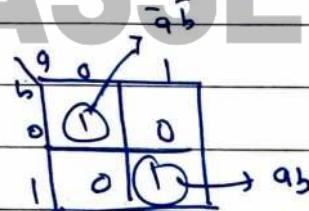
m.s.o.p = b

(as only a is changing, b is const.)

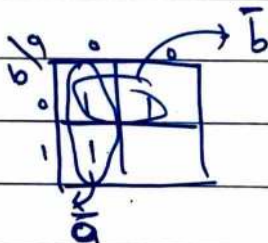


a

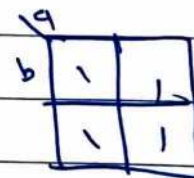
m.s.o.p = a



m.s.o.p = $ab + \bar{a}\bar{b}$



m.s.o.p = $\bar{a} + \bar{b}$



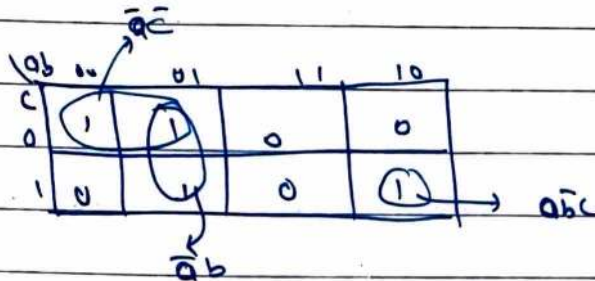
m.s.o.p = 1

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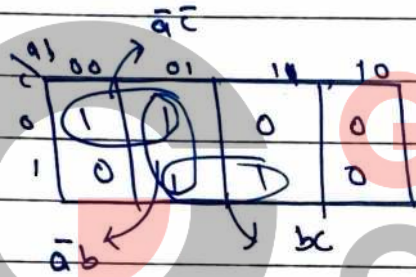
In cube with 2^m cells, m variables change so product term has $n-m$ literals where n is # var. in function.

Ex: 1



$$f = \bar{a}b + \bar{a}c + abc$$

Ex: 2



$$f = \bar{a}c + \bar{a}b + bc$$

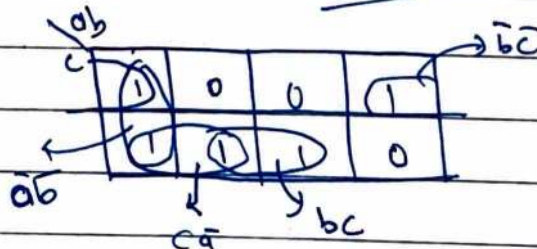
min: No.

$\bar{a}b$ cube is redundant.

$$f = \bar{a}c + bc$$

m.s.o.p.

Ex: 3



$$f = c\bar{a} + \bar{b}c + bc + \bar{a}b$$

(Not min)

$$f = \bar{a}b + c\bar{a} + bc + \bar{b}c \quad \text{OR} \quad c\bar{a} + bc + \bar{b}c$$

Both are m.s.o.p.

Not unique.

	a	b	f
m ₀	0	0	x
m ₁	0	1	y
m ₂	1	0	z
m ₃	1	1	w

=

a/b	0	1
0	x	z
1	y	w

=

a/b	0	1
0	x	y
1	z	w

Preferred.

Ques: Simplify bool. func. using kmap $f(a,b,c) = \sum(0,2,3,4,6)$

c/b	00	01	10	11
0	1	1	1	1
1	0	1	0	0

$$= \bar{a}b + \bar{c}$$

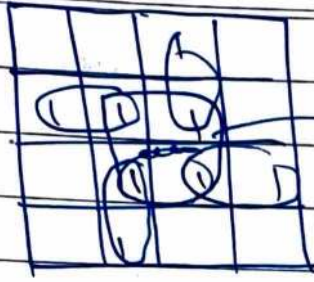
Rules for minimisation using k-map:

- First cover those 1 cells, by cubes that cannot be combine with other 1 cells.
- Cover 1 cells with min. # cubes such that each cube is as large as possible.
- Cube within another cube is not used in m.s.p.
- If there is more than one way of covering the map with minimal # cubes, select the cover with larger cubes (i.e. less # literals in m.s.p.)
- A cube contained in any combination of other cubes already selected in the cover is redundant by virtue of consensus theorem (i.e. $ab + \bar{a}c + bc$)

Redundant.

Minimal Exp. can be one among minimum SOP or mPOS or can be both min. exp.

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Redundant (larger cube)

⇒ Using K-map to get minimised pos expression

Here we cover zeroes in K-map with cubes to write sum terms & then product them.

Ques

Find mPOS of $f(a,b,c) = \sum (0,1,3,7)$

ab \ c	00	01	11	10
0	1	0	0	0
1	1	1	0	0

$\bar{a}+b$

Not changing
(Comp. for 1)

$$\text{mPOS} = (\bar{b}+c)(\bar{a}+b)$$

Ques If $f(a,b,c,d) = \sum (0,3,4,8,10,12)$, find mPOS?

cd \ ab	00	01	11	10
00	1	1	1	1
01	0	0	0	0
10	1	0	0	0
11	0	0	0	1

$\bar{d}+c$

$\bar{a}+\bar{d}$

$\bar{c}+\bar{a}+\bar{d}$

$\bar{b}+\bar{c}$

$$\text{mPOS} = (\bar{a}+\bar{d})(\bar{d}+c)(\bar{b}+\bar{c})(\bar{c}+\bar{a}+\bar{d})$$

→ There is no relation b/w #mPOS & #mSOP.

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Ques: If we make every cube of size 1, then which type of SOP will we derive?

⇒ Canonical SOP.

★ Functions with Don't Care Combinations: Partial function (Incompletely Specified fun)
or can occur for a bool. fun, sometimes

Some input combinations can NEVER occur, for these input comb we don't care about the fun. value. Such combinations are called Don't Care Combinations.

For Ex: $f(a,b) = 1$ if $a=1$

T.T.

a	b	f
0	0	X
0	1	X
1	0	1
1	1	1

Don't Care combinations as we don't care what happens when $a=0$.

f ₁	f ₂	f ₃	f ₄
0	0	1	1
0	1	0	1
1	1	1	1
1	1	1	1

4 fun. f in repr.

Each don't care can be specified as 0 or 1 (like joker card), a fun. with k don't cares corresponds to a class of 2^k distinct functions. Our aim for mSOP is to use any value of don't care which makes cube large.

Ex: $f(a,b) = \Sigma(1) + d(2,3)$

a	b	f
0	0	1
0	1	X
1	0	0
1	1	X

→ Treated as 1

→ Treated as 0

mSOP = 5

a	b	f
0	0	1
0	1	X
1	0	0
1	1	X

mPOS = 5

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Ex:2 Find msop for

$$f(w,x,y,z) = \sum m(0,2,4,5,8,14,15)$$

$$d(w,x,y,z) = \sum m(7,10,13)$$

w \ yz	00	01	11	10
00	1	1	0	1
01	0	1	X	0
11	0	X	1	0
10	1	0	1	X

Not req. / redundant.

$$msop = \bar{w}\bar{z} + \bar{w}xy + wxy$$

NOTE Largest cube may not be present in msop/mpoa.

Covering Functions:

f covers g only iff, whenever $g=1$, then $f=1$ OR whenever $f=0$ then $g=0$.

i.e.

g	f
1	1
0	0/1
0	0
1	0/1

f covers g here.

(Remember del.)

Ques: $f = a+b$ & $g = ab$. f cover g or g cover f ?
 $\Rightarrow$ When $f=1$ (0,1) $g=0$ so g don't cover f
 but when $g=1$ (1,1) $f=1$ so f covers g .

If f cover g & g cover f then $f=g$.

$$f \text{ covers } g \equiv g \text{ implies } f \equiv g=1 \rightarrow f=1 \equiv f=0 \rightarrow g=0$$

Covering fun. is defined from si Pov. (minterm)

Implies ka ulta covers

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$$f = n + \bar{y} + \bar{z}, \quad g = n + \bar{z}$$

here $f=0$ means $g=0$, so f covers g .

NOTE

$$\text{If } f(a,b,c) = \Sigma(0,5,7),$$

then covering fun. of f :

$$g(a,b,c) = \Sigma(0,5,7) + d(1,2,3,4,6)$$

OR

$$\text{If } f(a,b,c) = \Pi(0,5,7)$$

then covering fun. of f :

$$g(a,b,c) = \Sigma(1,2,3,4,6) + d(0,5,7).$$

Not valid for Π

→ mpos / m SOP for "covering fun of f " in " π " (check above note & make k-map).

→ Bigger product term implies its subset product term i.e. subset covers bigger product term.

Ex: $\alpha\beta\gamma \rightarrow \alpha\beta$ so $\alpha\beta$ covers $\alpha\beta\gamma$

→ Smaller ~~product~~ ^{sum} term implied its subset sum term i.e. subset term covers bigger sum term.

Ex: $\alpha + \beta\gamma \rightarrow \alpha + \beta\gamma$ so $\alpha + \beta\gamma$ covers $\alpha + \beta$

K-map is eff. for 54 var. only.

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Implicants (Imp.), Prime Implicants (PI), Essential Prime implicants (EPI)→ Every cube of 1's is an implicant.

Ex:

AB				
CD	1		1	
			1	
	1	1	1	1

Here we have

7 → 1 sized imp. / cube.

6 → 2 size imp.

1 → 4 sized imp.

total 14 implicants.

→ Best way to count implicants is grab each cell one by one & write all implicants from that cell & never see that cell again in any cube covering that cell as it's already covered.

→ Implicant is a product term which implies f . $\text{Imp} = 1 \rightarrow f = 1$ Obviously→ Prime Implicants: Cube of 1's which is as big as possible. (Subset of imp.)

Ex:

	1	1	1	1
		1	1	

 $\text{Imp} = 6 + 7 + 2 = 15$

Prime Imp: 2

Prime Imp is an imp. (from map pov) which is not fully contained within any other implicant.

→ Product term (as small as possible) which implies f , is PI. (don't have redundant literal)

Ex: 1

$$\# \text{ Imp} = 5. \quad (3+2)$$

Ex: 2

PI3 is not EPT

Ques:

$$\Sigma m(1, 2, 3, 5, 8, 9, 11, 15)$$

size 1 size 2 size 4
Imp: $2 + 2 + 1 = 17$.

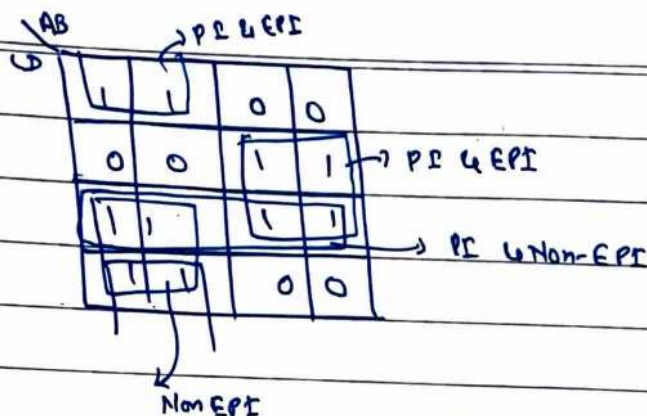
PI_2, PI_3, PT_4 e PI_5 m,
 EP_2 e PI_1 u,
 na EP_1 .

↳ will not be in msop also

While making cubes, always look at corners & folds

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Ex: 3

So mpos: $\bar{A}\bar{D} + AD + CD$

OR

$$\bar{A}\bar{D} + AD + \bar{A}C.$$

→ In map we have all EPI (must) & some non EPI (but all are PIs)

→ For don't cares, we consider them as 1 always to count implicants & also for PI we consider them 1. For EPI we don't consider it (same def. for EPI i.e. should uniquely cover at least one '1' cell).

Ex: 4

X	1		X
0	X	1	X

$$\text{Imp: } 6 + 6 = 12$$

$$\text{PI: } 6$$

$$\text{EPI: } 0.$$

* Cyclic k-map: Has exactly 2-maps. And both maps have 0 common term. Thus a is no EPI, every PI has same size & each 1 cell is covered by exactly 2 PIs.

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Ex: 1

	a	b		
c	1	1	0	1
	0	1	1	1

All condition satisfied.

mSOP1: $\bar{a}\bar{c} + cb + a\bar{b}$

mSOP2: $\bar{b}\bar{c} + \bar{a}b + ac$

Ex: 2

	a	b		
c	1	1	0	1
	0	1	1	1

✓

NOTE

Terms Imp, PI, EPI are defined only for SOP expressions (never cover 0 with them).

→ Some Author (non standard) define similar concept to implicant for 0 cells also called Implicate.

SOP
Implicant

Prime Implicant

E.P. "

POI
Implicate

Prime Implicate

Essential " "

Ex:

$\Sigma m(2, 3, 5, 7)$

	a	b		
c	0	1	0	0
	0	1	1	1

Implicate - $4 + 3 = 7$

Implicants - $4 + 3 = 7$

Dual is Complement of a function

$$f^d(a, b, \bar{a}, \bar{b}, +, \cdot, 1, 0)$$

$$\downarrow$$

$$f(a, b, \bar{a}, \bar{b}, +, \cdot, 0, 1)$$

$$\downarrow$$

$$\bar{f}(\bar{a}, \bar{b}, a, b, +, \cdot, 0, 1)$$

So in dual of function, variables remain unchanged. (Also Not)

Ex: Dual of eqn $(a)(a+b)=a$ is $(a)+(ab)=a$

Dual of $(\bar{x}+\bar{y})=\bar{xy}$ is $(\bar{x}\bar{y})=\overline{(x+y)}$

$$f^d(a, b, \bar{a}, \bar{b}, +, \cdot, 1, 0)$$

$$\downarrow$$

$$\bar{f}(\bar{a}, \bar{b}, a, b, +, \cdot, 1, 0)$$

→ So only literals change when $f^d \leftrightarrow \bar{f}$.

Ex if $f^d = a+b$

then, $\bar{f} = \bar{a} + \bar{b}$

or $\bar{f} = \underline{\underline{ab}}$

→ So, for $f^d(0,0) = \bar{f}(1,1)$

$f^d(1,0) = \bar{f}(0,1)$

so on.

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Ex:

a	b	$a+b$ $f(a,b)$	$\bar{f} = a\bar{b}$	$f^d = \bar{a}b$
0	0	1	0	0 ($\bar{f}(1,1)$)
0	1	1	0	1 ($\bar{f}(1,0)$)
1	0	0	1	0 ($\bar{f}(0,1)$)
1	1	1	0	0 ($\bar{f}(0,0)$)

So:

a	b	f	$\bar{f}$	f^d
0	0	x	$\bar{x}$	3
0	1	y	$\bar{y}$	2
1	0	z	$\bar{z}$	1
1	1	w	$\bar{w}$	4

So in dual we basically ^{rev.} the complemented row. (if 1 in std. form).

Ques: $F = (\bar{y}w + x)y + \bar{z}$

Verify $f^d(0,1,0,0,1) = \bar{f}(1,0,1,1,0)$

$$\begin{aligned}
 f^d &= (((\bar{y} + w) \cdot x) + y) \cdot \bar{z} \\
 &= (((1 + 1) \cdot 0) + 0) \cdot 0 \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \bar{f} &= (((y + \bar{w}) \cdot \bar{x}) + y) \cdot z \\
 &= (((1 + 1) \cdot 0) + 0) \cdot 0 \\
 &= \underline{\underline{0}}
 \end{aligned}$$

→ In std. T.T. we write input comb. in order of their decimal value.

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Ques: How to find dual if T.T. is given for dual.
 ⇒ From T.T., find $\bar{f}$ & then f 's expression
 and from expression of f , find dual.

Ex:

a	b	f^d	$\bar{f}$	f
0	0	0	1	0
0	1	1	1	0
1	0	1	1	0
1	1	0	0	1

$$f = a \cdot b$$

$$\underline{\underline{f^d = a + b}}$$

Note that $f^d(t) = \bar{f}(2^n - 1 - t)$ for f over n variables
 OR
 $\bar{f}(t) = f^d(2^n - 1 - t)$

If $f(x) = 0$ then $\bar{f}(x) = 1$

if $\bar{f}(x) = 1$ then $f^d(2^n - 1 - x) = 1$

if $f(x) = 0$ then $f^d(2^n - 1 - x) = 1$ ✓

★ Neutral Boolean function: A bool. fun. 'f' over n var. is neutral iff $f=1$ for half of input combinations and $f=0$ for remaining half.

Ex: $f = a \oplus b$, $a \odot b$

neutral functions =

$$2^n \cdot C_{2^{n-1}-1}^{2^n-1}$$

where n var. function

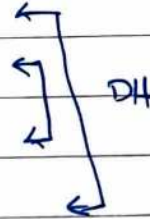
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$\Rightarrow f = \bar{f} \rightarrow \text{Never}$, $f \neq \bar{f} \rightarrow \text{Always}$, $f = f' = \text{some times}$.

* Self Dual functions: Function f is self dual iff $f = f'$.

a	b	f	f'
0	0	a_0	$\bar{a}_0$
0	1	a_1	$\bar{a}_1$
1	0	a_2	$\bar{a}_2$
1	1	a_3	$\bar{a}_3$



So, for self dual fun $a_0 = \bar{a}_3$, $a_1 = \bar{a}_2$, $a_2 = \bar{a}_1$ & $a_3 = \bar{a}_0$

So $a_0 \neq a_3$, $a_1 \neq a_2$, $a_2 \neq a_1$ & $a_3 \neq a_0$

Self Dual function $\rightarrow$ Neutral

Ex:

a	b	f
0	0	0
0	1	1
1	0	1
1	1	0

Neutral but not self dual.

Half of the ^{rows} elements have 2 choices so
 # self dual fun: $\frac{(2)^{2^n}}{2}$ for n variables

* In self dual fun.

$$f(n) \neq (2^n - 1 - n)$$

Defined over

$$\text{for } n \leq 2^n - 1$$

Ques: $f = \{0, 1, 2, 3, 6\}$ is f self dual?

Ans: $f(0) \neq f(7)$
 $f(1) \neq f(6)$
 $f(2) \neq f(5)$
 $f(3) \neq f(4)$

Exactly 1 among pair is 1.

Another special fun. is those fun whose comp. is same as dual (No specific name).

→ Orthogonal Function? (No) } Misconception.
 → No such word

Ex:

a	b	f	$\bar{f}$	f^d
0	0	0	1	1
0	1	1	0	0
1	0	1	0	0
1	1	0	1	1

So $\bar{f} = f^d$.

Ques: Is such function neutral always?

Ans: No. Ex:

a	b	f	$\bar{f}$	f^d
0	0	0	1	1
0	1	0	1	1
1	0	0	1	1
1	1	0	1	1

same

$\bar{f} = f^d$ but not neutral.

for such fun $f(n) = f(2^n - 1 - n)$ for $0 \leq n \leq 2^n - 1$ } Palindromic string.

such fun = 2^{n-1}

→ Each row has 2 choices

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★ XOR, XNOR functions & their properties:

NOTE Cancellation law does not hold in boolean algebra.

Ex NOR is also known as Eq. opn / Coincidence opⁿ. It is same as bimplication of prop. logics.

$$a \oplus b = \bar{a} \odot b = a \odot \bar{b} = \overline{a \odot b} = \bar{a} \oplus \bar{b}$$

Think of what you are changing & how is it affecting us

★ XOR with more than 2 inputs: Mod 2 sum

It is comm & associative.

→ So XOR is 1 if odd no. of 1's in i/p.

$$1 \oplus 1 \oplus 1 \oplus \dots = 1 \text{ (any 0)}$$

$$a \oplus a \oplus a \oplus a = 0$$

↳ bcz odd is not possible.

★ XNOR is 1 if even no. of 1's in i/p.

$$\begin{array}{ccccccc} 0 \odot 0 \odot 0 \odot 0 \odot 0 & = & 1 \\ 0 \oplus 0 \oplus 0 \oplus 0 & = & 0 \end{array} \text{ (any 1s)}$$

⇒ XOR is odd 1 checker & XNOR is even 0 checker.

→ XOR is equal to XNOR when odd no. of i/p are considered

Ex: $a \oplus b \oplus c = a \odot b \odot c$

↳ always bcz LHS is 1 when 1 or 3 1's
RHS is 1 when 0 or 2 1's
Same always.

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★ Positive Logic System & Negative Logic System:

Binary logic is always 0s & 1s but electricity don't come as 0 or 1, it comes in volts. So we assign volts to 0's & 1's.

In positive logic we assign 0s to low voltage & 1's to high voltage.

In negative logic we assign 0s to high voltage and 1's to low voltage.

Ex:

Digital Circuit			+ive Logic System			-ive Logic System		
X	Y	F(X,Y)	X	Y	F	X	Y	F
L	L	L	0	0	0	1	1	1
L	H	H	0	1	1	1	0	0
H	L	H	1	0	1	0	1	0
H	H	H	1	1	0	0	0	0

∴ +ive → -ive logic

AND → OR,

OR → AND,

1 → 0 (High)

0 → 1 (Low)

A → A

Conclusion is +ive logic system & -ive logic system are duals of each other.

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I would love to know if you liked my notes. Your feedback and appreciation are invaluable, so don't hesitate to share your thoughts. Click [here](#) to connect with me on LinkedIn or you can find me as @agrawalkaran.

Wishing you success on your journey ahead!

- Karan Agrawal